Pattern Matching

Concepts of Programming Languages

CAS CS 320

Outline

- Discuss **pattern matching**
- This means giving a syntax, typing, and semantics
- This also means defining **patterns**, and describing what matching actually means

demo

Pattern Matching

Pattern Matching (High Level)

```
let rec sum l =
  match l with
  | [] -> 0
  | x :: xs -> x + sum xs
```

Values come in different "shapes"

We want to do things based on the different shapes

Matching allows us to condition on the shape of the value

```
sum (1 :: 2 :: [])
```

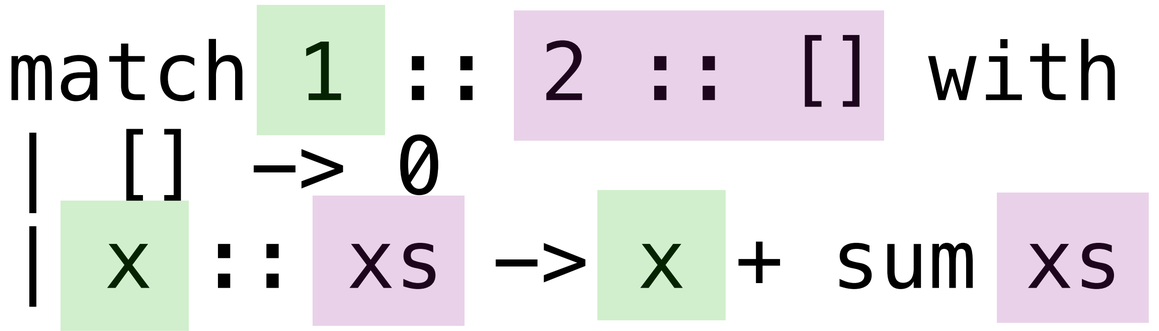
```
match 1 :: 2 :: [] with
| [] -> 0
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```

```
match 1 :: 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
```

failed match

```
match 1 :: 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
```

successful match



create new bindings

```
1 + sum (2 :: [])
```

compute branch with new bindings

```
 1 +
match 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
```

rinse and repeat

```
1 +
match 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
```

failed match

```
1 +
match 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
```

successful match

```
1 +
match 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
```

create new bindings

```
1 + (2 + sum [])
```

compute branch with new bindings

```
1 + (2 +
match [] with
| [] -> 0
| x :: xs -> x + sum xs)
```

rinse and repeat

```
1 + (2 +
match [] with
| [] -> 0
| x :: xs -> x + sum xs)
```

successful match

```
1 + (2 +
match [] with
| [] -> 0
| x :: xs -> x + sum xs)
```

no new bindings

```
1 + (2 + 0)
```

compute branch

3

fin

Making Things Formal

```
let rec sum l =
  match l with
  | [] -> 0
  | x :: xs -> x + sum xs
```

What do match-expressions look like? **(syntax)**

What is a valid match-expression? **(typing)**

How do match-expressions behave? **(semantics)**

Syntax

Match Syntax

```
match l with
| [] → e1
| x::xs → e2
```

⟨e⟩ ::= match ⟨e⟩ with ⟨a⟩ { ⟨a⟩ }

⟨a⟩ ::= | ⟨e⟩ → ⟨e⟩

zero or more

Match Syntax (Example)

```
match 2 + 4 with
| 2 + 4 -> 2 * 4
| _ -> 0
```

Can we do this in OCaml?

Match Syntax (Again)

$e ::= n \mid \lambda x.\, e \mid e\,e \mid \texttt{let } x = e \texttt{ in } e$
$\mid e \oplus e \mid \texttt{true} \mid \texttt{false}$
$\mid [\,] \mid e :: e$

$\oplus ::= + \mid - \mid * \mid ::$

$e ::= \texttt{match } e \texttt{ with } a^+$

$a ::= \mid p \to e$

$p ::= n \mid \texttt{true} \mid \texttt{false}$
$\mid [\,] \mid p :: p$
$\mid x \mid _$

variable wildcard

Patterns (High Level)

```
match x with
| 1 -> e1
| 2 -> e2
| _ -> e3
```

A pattern describes the "shape" of a **value**

But we don't *just* care about the "shape" we also want to **do stuff** with the pieces of the data

Patterns are syntactic representations of values with names

Shallow vs. Deep Pattern Matching

```
let third l =
  match l with
  | _ :: _ :: x :: _ -> Some x
  | _ -> None
```

We're interested implementing *deep* pattern matching, which allows for *nested* patterns

We can also implement *shallow* pattern matching ADTs (which are sometimes called *eliminators*)

Shallow Pattern Matching for Lists

```
<expr> ::= match <expr> with
           | [] -> <expr>
           | <var> :: <var> -> <expr>
```

We can hardcode the possible patterns

This works, but is tedious

Shallow Pattern Matching for Lists

```
let third l =
  match l with
  | [] -> None
  | _ :: l -> (
      match l with
      | [] -> None
      | _ :: l -> (
          match l with
          | [] -> None
          | x :: _ -> Some x
        )
    )
```

Wildcards

```
let third l =
  match l with
  | _ :: _ :: x :: _ -> Some x
  | _ -> None
```

Like OCaml, we'll treat wildcards as distinct from variables

Pattern Syntax

Match Syntax (Again, Again)

Typing

Questions

What makes a valid match expression?

When should I get a type error?

(1) match arms output are the same

(2) patterns but be the "right type"

Match Typing

problem: don't know about pattern ids.

$$\frac{\Gamma \vdash e_0 : T_0 \qquad \boxed{\Gamma \vdash p_1 : T_0 \quad \Gamma \vdash e_1 : T \quad \cdots \quad \Gamma \vdash p_k : T_0 \quad \Gamma \vdash e_k : T}}{\Gamma \vdash \text{match } e_0 \text{ with } \mid p_1 \to e_1 \cdots \mid p_k \to e_k : T} \text{ (match)}$$

Pattern Typing (High Level)

```
match 1 :: 2 :: 3 with
| [] -> 0
| true -> 2
| x :: _ -> x
```

type error

Pattern Typing (High Level)

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match 1 :: 2 :: 3 with
| [] -> 0
| true -> 2
| x :: _ -> x
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type error

Patterns are *typed*. We can check for "reasonableness" of patterns

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(In fact, we can check if we have exactly enough patterns, this is *exhaustiveness checking*)

Pattern Typing (High Level)

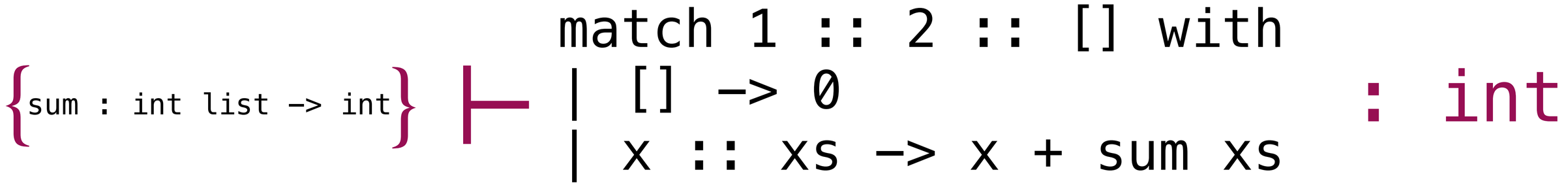
```
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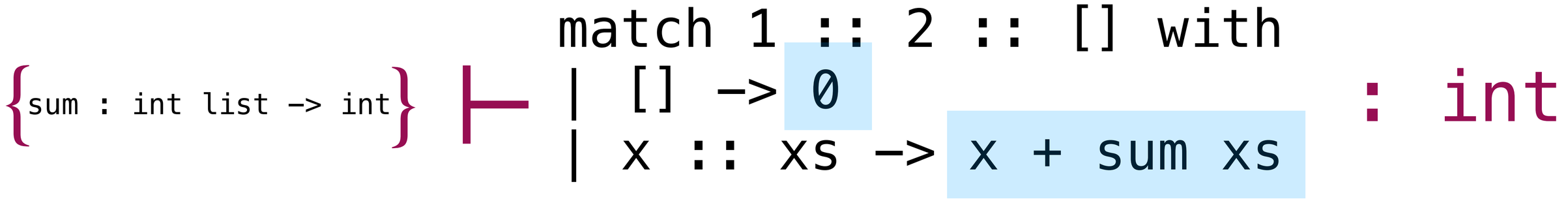
type error

Patterns are *typed*. We can check for "reasonableness" of patterns

(In fact, we can check if we have exactly enough patterns, this is *exhaustiveness checking*)

Key point: patterns *generate* variable declarations


{sum : int list -> int} ⊢
match 1 :: 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
: int



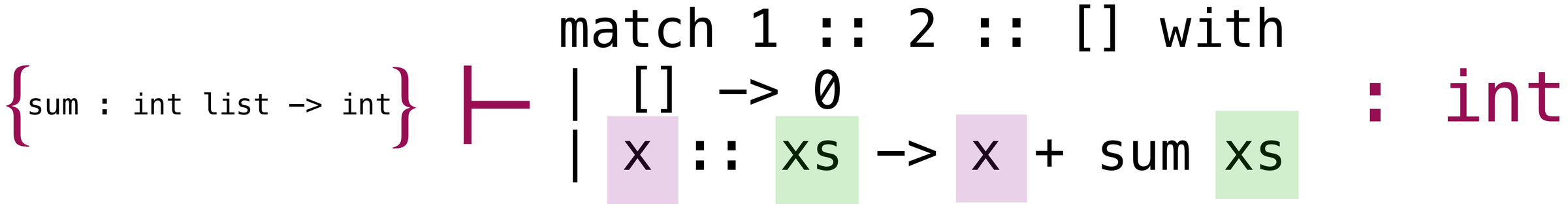
check both branches are ints

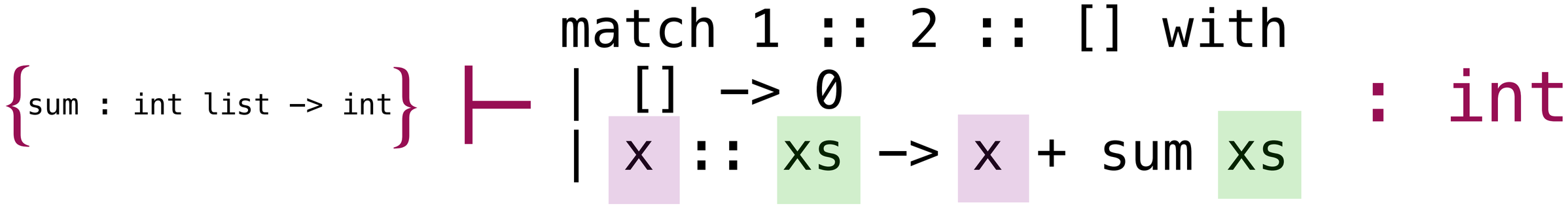
$$\{\texttt{sum : int list -> int}\} \vdash 0 : \texttt{int}$$


{sum : int list -> int} ⊢ x + sum xs : int

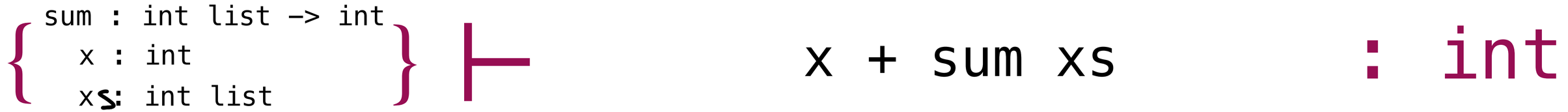


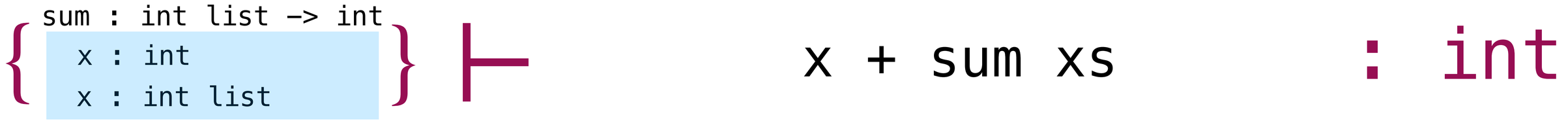
We don't know what x and xs are


{sum : int list -> int}
⊢
match 1 :: 2 :: [] with
| [] -> 0
| x :: xs -> x + sum xs
: int



the pattern generates new declarations


sum : int list -> int
x : int
xs : int list
x + sum xs
: int



How do we generate these declarations?

Pattern Typing Judgments

$$p : \tau \Rightarrow \Gamma$$

"The pattern p is of type τ and generates the variable declarations Γ"

(Our derivations now have multiple kinds of judgments)

Pattern Typing

$p : \tau \Rightarrow \Gamma$

$e ::= n \mid \lambda x.e \mid ee \mid \text{let } x = e \text{ in } e$
$\mid e \oplus e \mid \text{true} \mid \text{false}$
$\mid [\,]$

$\oplus ::= + \mid - \mid * \mid ::$

$e ::= \text{match } e \text{ with } a^+$

$a ::= \mid p \to e$

$p ::= n \mid \text{true} \mid \text{false}$
$\mid [\,] \mid p :: p$
$\mid x \mid _$

x: variable, $_$: wildcard

$$\frac{n \in \mathbb{Z}}{n : \text{int} \Rightarrow \emptyset}$$

$$\frac{}{_ : \tau \Rightarrow \emptyset}$$

$$\frac{}{[\,] : \tau \text{ list} \Rightarrow \emptyset}$$

$$\frac{}{\text{true} : \text{bool} \Rightarrow \emptyset}$$

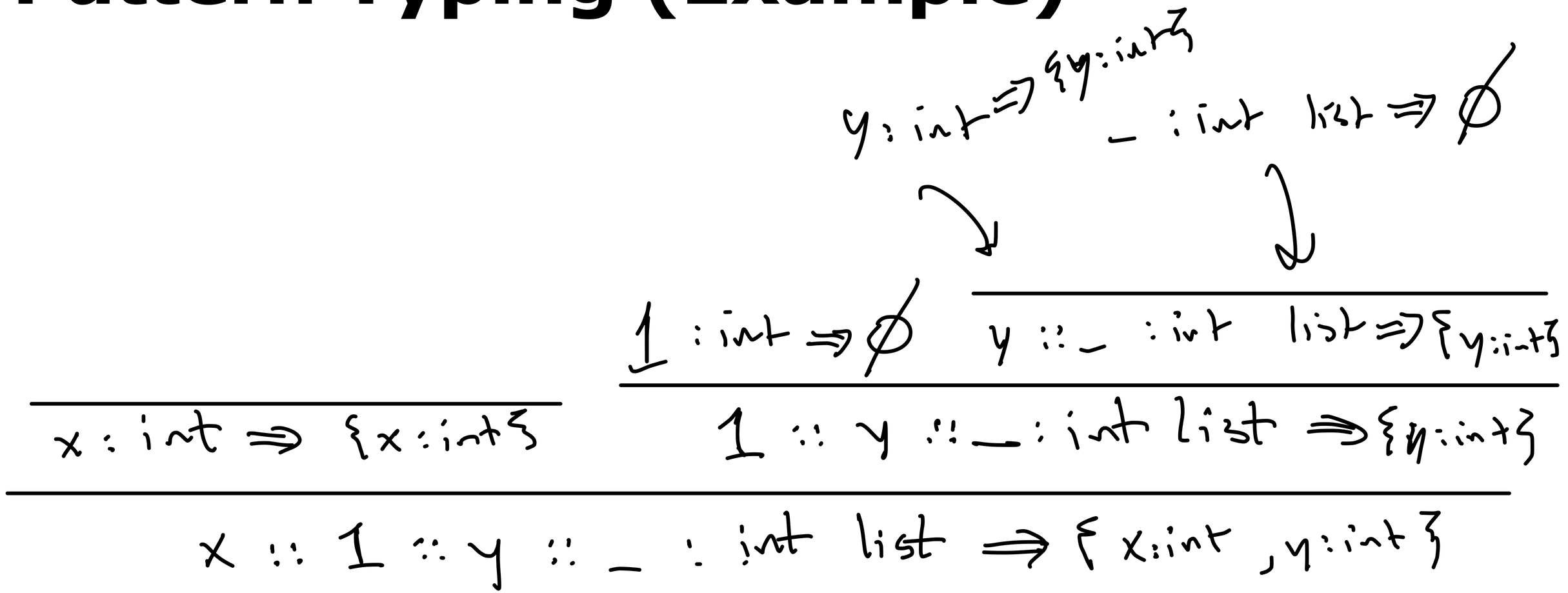
$$\frac{}{x : \tau \Rightarrow \{x : \tau\}}$$

Γ_1, Γ_2 are disjoint

$$\frac{}{\text{false} : \text{bool} \Rightarrow \emptyset}$$

$$\frac{p_1 : \tau \Rightarrow \Gamma_1 \qquad p_2 : \tau \text{ list} \Rightarrow \Gamma_2}{p_1 :: p_2 : \tau \text{ list} \Rightarrow \Gamma_1, \Gamma_2}$$

Pattern Typing (Example)



Match Typing (Again)

Match Typing (Example)

Semantics

Questions

What does "matching" mean?

How do we find a match?

What do we do when we find a match?

What if we don't find a match?

Matching Semantics

Pattern Matching (High Level)

```
let rec sum l =
  match l with
  | [] -> 0
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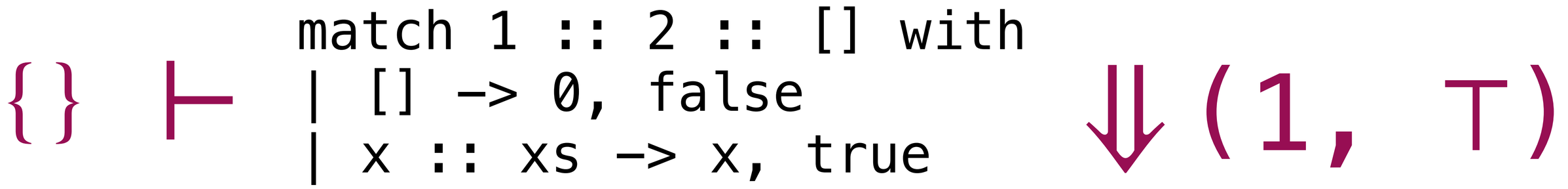
Since patterns "look" like values, we just need to describe when two things "look alike"

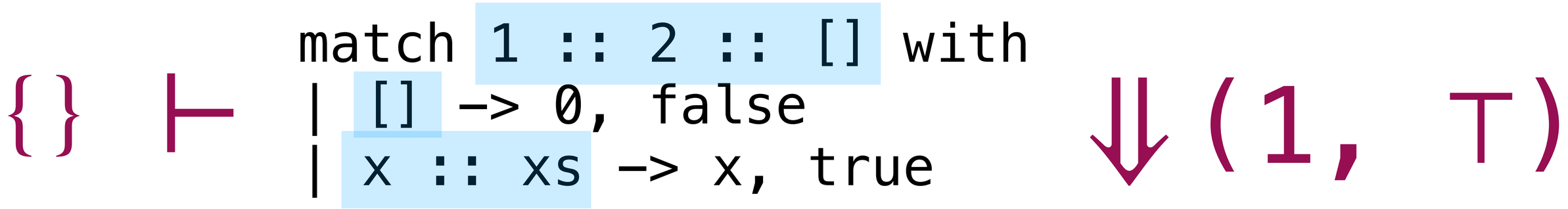
Pattern Matching (High Level)

```
let rec sum l =
  match l with
  | [] -> 0
  | x :: xs -> x + sum xs
```

Since patterns "look" like values, we just need to describe when two things "look alike"

Reminder: matching generates variable bindings


{} ⊢
match 1 :: 2 :: [] with
| [] -> 0, false
| x :: xs -> x, true
⇓(1, ⊤)



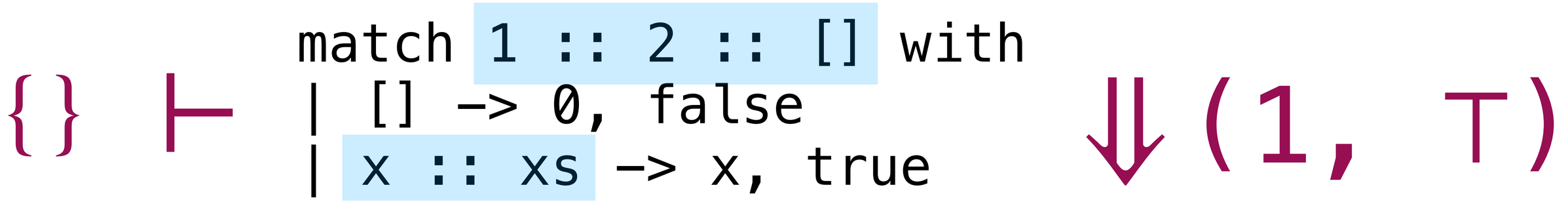
find a branch that the value of the expression matches

$\{\} \vdash$

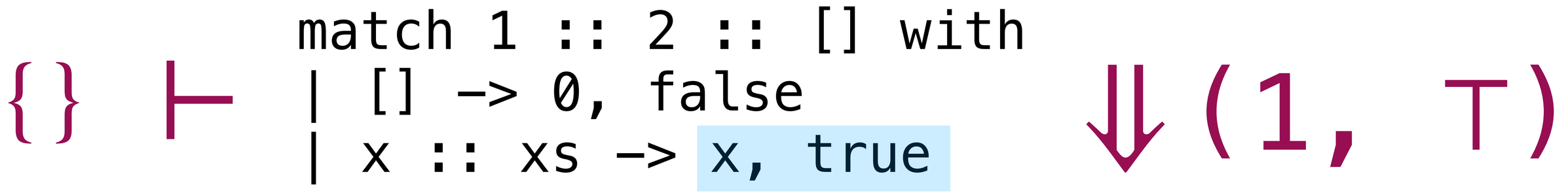
```
match 1 :: 2 :: [] with
| [] -> 0, false
| x :: xs -> x, true
```

$\Downarrow (1, \top)$

failed pattern match

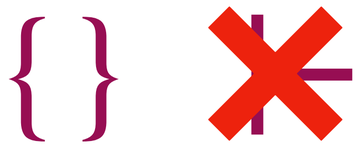


successful pattern match



evaluate success branch

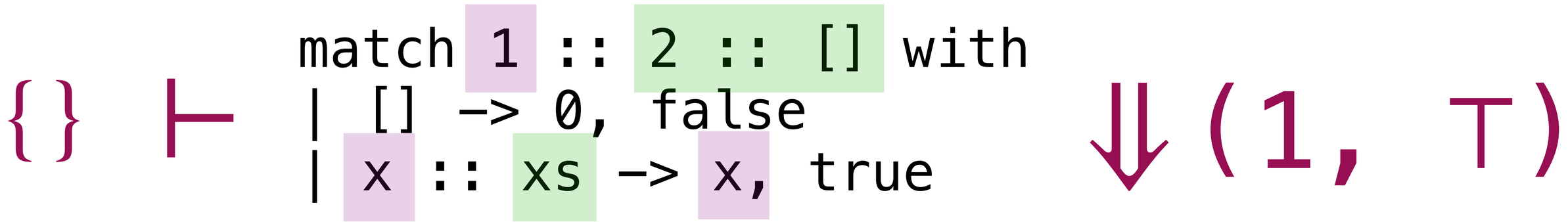
$$\{\} \vdash \texttt{x, true} \Downarrow (1, \top)$$



x, true



We don't know what x is



the pattern generates new bindings

$$\{\texttt{x} \mapsto \texttt{1}, \texttt{xs} \mapsto \texttt{[2; 3]}\} \vdash \texttt{x, true} \Downarrow (1, \top)$$

$$\{x \mapsto 1, xs \mapsto [2; 3]\} \vdash x, \texttt{true} \Downarrow (1, \top)$$

How do we generate these declarations?

Pattern Matching Judgment

$$v \text{ matches } p \Rightarrow \mathscr{E}$$

"The value v matches the pattern p and generates the bindings $\mathscr{E}$"

Pattern Matching

Pattern Matching (Example)

Match Semantics

Match Semantics (Example)

Summary

We can formally define the syntax, typing and semantics of **match expressions**

Deep pattern matching requires the introduction of **patterns**